

Pterosaur integumentary structures with complex feather-like branching

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Pterosaurs were the first vertebrates to achieve true flapping flight, but in the

absence of living representatives, many questions concerning their biology and

lifestyle remain unresolved. Pycnofibres, the integumentary coverings of

pterosaurs, are particularly enigmatic: although many reconstructions depict

fur-like coverings composed of pycnofibres, their affinities and function are not fully understood. Here we report the preservation in two anurognathid pterosaur specimens of morphologically diverse pycnofibres that show diagnostic features of feathers, including non-vaned grouped filaments and bilaterally branched filaments, hitherto considered unique to maniraptoran dinosaurs, and preserved melanosomes with diverse geometries. These findings could imply that feathers had deep evolutionary origins in ancestral archosaurs, or that these structures arose independently in pterosaurs. The presence of feather-like structures suggests that anurognathids, and potentially other pterosaurs, possessed a dense filamentous covering that likely functioned in thermoregulation, tactile sensing, signalling, and aerodynamics.

Feathers are the most complex integumentary appendages¹. Most feathers in modern birds possess an axial shaft from which branch lateral barbs and barbules. Much is known about the anatomy, developmental biology, and genomic regulation of these structures, but their deep evolutionary origin is controversial²⁻⁴. Feathers and feather-like integumentary structures have been reported in many theropod dinosaurs (including birds)^{3,5} and ornithischians such as *Psittacosaurus*⁶, *Tianyulong*⁷, and *Kulindadromeus*⁸. Feather-like or hair-like structures, termed pycnofibres⁹, have also been reported in several pterosaur specimens⁹⁻¹³, but their nature is not resolved.

Here we report remarkably well-preserved pycnofibres in two anurognathid pterosaurs and demonstrate, using evidence from morphology, chemistry and

macroevolutionary analyses, that the preserved pycnofibres bear key features of feathers: monofilaments, two types of non-vaned grouped filaments, bilaterally branched filaments that were previously considered unique to maniraptoran dinosaurs, and preserved melanosomes with diverse geometries. Both specimens studied are from the Middle–Late Jurassic Yanliao Biota (ca. 165–160 Mya¹⁴). NJU–57003 (Nanjing University) is a newly excavated specimen from the Mutoudeng locality and CAGS–Z070 (Institute of Geology, Chinese Academy of Geological Sciences), which has been noted previously¹³, is from the Daohugou locality. Both specimens are near-complete and well-articulated, with extensive soft tissues (Figs. 1 and 2, and Supplementary Figs. 1–5). Both specimens are identified as anurognathids¹⁷ (see Supplementary text for osteological descriptions).

Preserved soft tissues include structural fibres (actinofibrils) and pycnofibres. Structural fibres, common in the pterosaur wing membrane^{9,12,18}, are observed only in the posterior portion of the uropatagium in CAGS–Z070 (Fig. 1**o–p**). As reported elsewhere, they are parallel to subparallel and closely packed. Individual fibres are 0.08–0.11 mm wide (ca. 5 fibres per mm) and at least 1.9 mm long. Pycnofibres are preserved extensively in both pterosaur specimens (especially CAGS–Z070; Figs. 1 and 2, and Supplementary Figs. 1, 4 and 5) and are discriminated from structural fibres based on their curved morphology and overlapping arrangement. In the posterior portion of the uropatagium in CAGS–Z070, pycnofibres co-occur with structural fibres; oblique intersections reflect superposition of these features during decay (Fig. 1**o–p**).

68 Pycnofibres are categorized here into four types. Type 1 occurs around the head,
 69 neck, shoulder, torso, all four limbs and tail of both specimens (Figs. 1**c–e**, **o–p**, 2**b–c**
 70 and **f**). It comprises curved monofilaments that are 3.5–12.8 mm long and 70–430 μm
 71 wide. Some short, distally tapering examples discriminate between dark-toned lateral
 72 margins and light-toned axial regions, especially near the filament base where the
 73 light-toned axis is wider, suggesting a tube-like morphology (Fig. 1**c–e**). Type 2 is
 74 preserved in the neck, proximal forelimb, plantar metatarsus and proximal tail regions
 75 of CAGS–Z070. It consists of bundles of curved filaments of similar length that
 76 appear to form brush-like structures at the distal ends of thicker filaments (2.0–13.8
 77 mm long and 80–180 μm wide) (Fig. 1**f–h**). The latter may represent individual thick
 78 filaments or fused proximal regions of thinner distal filaments. Type 3 occurs around
 79 the head of CAGS–Z070. It comprises straight to slightly curved, distally tapered,
 80 central filaments (4.5–7.0 mm long and 50–450 μm wide) with short lateral branches
 81 that diverge from the central filament near the midpoint (Fig. 1**i–k**). There are five
 82 Type 3 filaments identified on the head, next to five similar filaments likely of the
 83 same nature but obscured by overlapping filaments (Supplementary Fig. 5**b**). Type 4
 84 occurs on the wing membrane of both specimens. It comprises tufts of curved
 85 filaments (2.5–8.0 mm long and 70–130 μm wide) that diverge proximally (Figs. 1**l–n**
 86 and 2**d–e**), in contrast to the clear separation between Type 1 filaments (Fig. 1**o–p**).

87 Filamentous integumentary structures in extant and fossil vertebrates commonly
 88 contain melanin-bearing organelles (melanosomes). Scanning electron microscopy
 89 (SEM) of the filamentous structures of NJU–57003 reveals densely packed

microbodies $0.70 \pm 0.11 \mu\text{m}$ long and $0.32 \pm 0.05 \mu\text{m}$ wide (Fig. 2g–h, Supplementary Figs. 4a–f, 6 and 7, and Supplementary Table 2). As with most melanosome-rich fossil feathers^{19–21}, energy dispersive X-ray spectroscopy (EDS) spectra of the filaments are dominated by a major peak for carbon (Supplementary Fig. 8). These carbonaceous microbodies resemble fossil melanosomes in terms of their geometry, dense packing, parallel alignment relative to the long axis of the integumentary structure (i.e. barbules in Paraves), and preservation within the matrix of the filament (see Supplementary text). Most of the microbodies are oblate and morphologically similar to those that are usually interpreted as phaeomelanosomes in fossils¹⁹ (Fig. 2h). Rod-shaped examples, usually interpreted as eumelanosomes in fossils¹⁹ (Fig. 2g), are rare.

Fourier transform infrared spectroscopy (FTIR) of samples of pterosaur filaments shows four major peaks unique to the filaments (Fig. 2i). These peaks are consistent with the absorption regions of amide I at ca. 1650 cm^{-1} (principally the C=O asymmetric stretching vibration with some C–N bending), amide II at ca. 1540 cm^{-1} (a combination of N–H in-plane bending and C–N and C–C stretching as in indole and pyrrole in melanin and amino acids), and aliphatic C–H stretching at 2850 cm^{-1} and 2918 cm^{-1} ²². These peaks also occur in spectra obtained from extant feathers^{21,23}, fossil feathers of the paravian *Anchiornis*²⁰, and melanosomes isolated from human hair²⁴. Further, spectra of the pterosaur filaments more closely resemble those of pheomelanin-rich red human hair in the stronger absorption regions at ca. 2850 cm^{-1} and 2918 cm^{-1} and higher resolution in the region ca. $1500\text{--}1700 \text{ cm}^{-1}$ than those

from eumelanin-rich black human hair and the ink sac of cuttlefish²⁴. This, together with the SEM results, suggests that the densely packed microbodies in the pterosaur filaments are preserved melanosomes. The amide I peak at 1650 cm^{-1} is more consistent with α -keratin (characteristic of extant mammal hair²⁵) than β -keratin (the primary keratin in extant avian feathers^{22,26}). This signal may be original or diagenetic; the molecular configuration of keratin²⁶ and other proteins²⁷ can alter under mechanical stress and changes in hydration levels.

The ultrastructural and chemical features of the pterosaur filaments confirm that they are hair-like or feather-like integumentary structures. The four types of filaments described here show distinct distributions and morphologies. They are separated clearly from the sedimentary matrix by sharp boundaries (Supplementary Fig. 4g–i). There is no evidence that one or more filament type(s) were generated taphonomically, e.g. through selective degradation or fossilization, or superimposition of filaments. For instance, although Type 1 and 4 filaments occur widely in both specimens, Type 4 occurs only in the wings, while Type 1 occupies the remaining body regions. Type 1 filaments are thus not degraded products of Type 4, and Type 4 filaments do not represent superimposed clusters of Type 1 filaments. Filament types 2 and 3 occur only in CAGS–Z070. Type 3 occurs only in the facial area and is associated with Type 1, where Types 2 and 4 are not evident. Type 3 filaments are thus not degraded Type 2 or 4 filaments. Central filaments of Type 3 are morphologically identical to the short, distally tapering filaments of Type 1, but the branching filaments are much thinner ($< 40\text{ }\mu\text{m}$ (Type 3) versus $>70\text{ }\mu\text{m}$ (Type 1) wide) and shorter ($< 0.6\text{ mm}$ vs. $> 3.5\text{ mm}$

134 long) than the latter. The branching filaments are thus unlikely to reflect
135 superimposition of clusters of Type 1 filaments. In contrast, the distal ends of Type 2
136 filaments are similar, and have a similar distribution pattern to, Type 1 filaments. An
137 alternative interpretation, that Type 2 filaments might represent superimposition of
138 Type 1 filaments at their proximal ends, is unlikely (see detailed discussion in
139 Supplementary text).

140 Feathers and feather-like integumentary structures have been reported in
141 non-avian dinosaurs, although debate continues about their true nature². These
142 structures have been ascribed to several morphotypes, some absent in living birds^{3,5},
143 and provide a basis to analyse the evolutionary significance of pterosaur pycnofibres.
144 The pterosaur Type 1 filaments resemble monofilaments in the ornithischian
145 dinosaurs *Tianyulong* and *Psittacosaurus* and the coelurosaur *Beipiaosaurus*:
146 unbranched, cylindrical structures with a midline groove that widens towards the base
147 (presumed in *Beipiaosaurus*)^{3,5}. The pterosaur Type 2 filaments resemble the
148 brush-like bundles of filaments in the coelurosaurs *Epidexipteryx* and *Yi*^{3,5,28}: both
149 comprise parallel filaments that unite proximally. The morphology and circum-cranial
150 distribution of pterosaur Type 3 filaments resemble bristles in modern birds¹, but
151 surprisingly do not correspond to any reported morphotype in non-avian dinosaurs.
152 The Type 3 filaments recall bilaterally branched filaments in *Sinornithosaurus*,
153 *Anchiornis*, and *Dilong*, but the latter filaments branch throughout their length rather
154 than halfway along the central filament(s), as in the pterosaur structure^{3,5}. The
155 pterosaur Type 4 filaments are identical to the radially branched, downy feather-like

156 morphotype found widely in coelurosaurs such as *Sinornithosaurus*, *Beipiaosaurus*,
157 *Protarchaeopteryx*, *Caudipteryx*, and *Dilong*^{3,5}.

158 The filamentous integumentary structures in our anurognathid pterosaurs are thus
159 remarkably similar to feathers and feather-like structures in non-avian dinosaurs.
160 Intriguingly, cylindrical (Type 1), radially symmetrical branched (Types 2 and 4) and
161 bilaterally symmetrical branched (Type 3) filaments clearly coexisted in individual
162 animals; these structures may represent transitional forms in the evolution of feathers,
163 as revealed by developmental studies^{3,5}.

164 These new findings warrant revision of the origin of complex feather-like
165 branching integumentary structures from Dinosauria to Avemetatarsalia, the wider
166 clade that includes dinosaurs, pterosaurs, and close relatives^{4,29}. The early
167 evolutionary history of bird feathers and homologous structures in dinosaurs, and the
168 multiple complex pycnofibres of pterosaurs, is enigmatic. A previous study concluded
169 that the common ancestor of these clades bore scales and not filamentous
170 integumentary appendages², but this result emerged only when the filaments of
171 pterosaurs were coded as non-homologous with those of dinosaurs. There are no
172 morphological criteria, however, for such a determination. The presence of multiple
173 pycnofibre types and their morphological, ultrastructural and chemical similarity to
174 feathers and feather-like structures in various dinosaurian clades, confirms their likely
175 homology with filamentous structures in non-avian dinosaurs and birds. Comparative
176 phylogenetic analysis produces equivocal results: maximum likelihood modelling of
177 plausible ancestral states, against various combinations of branch length and character

transition models (Supplementary text and Supplementary Fig. 9, Table 3), reveals various potential solutions. The statistically most likely result (Fig. 3 and Supplementary Table 3, highest log-likelihood value) shows that the avemetatarsalian ancestors of dinosaurs and pterosaurs possessed integumentary filaments, with highest likelihood of possessing monofilaments; tufts of filaments, and, especially, brush-type filaments, are less likely ancestral states. This confirms that feather-like structures arose in the Early or Middle Triassic. The alternative tree for Dinosauria, with Ornithischia and Theropoda paired as Ornithoscelida³⁰, produces an identical result.

We present these modelling data with caution, however, for two reasons: (1) the tree rooting method can influence the result (Supplementary Table 3), favouring results in which either scales are the basal condition or where non-theropod feather-like structures and feathers evolved independently (Supplementary Figure 9, Table 3), and (2) there is no adequate way to model probabilities of evolution of all six feather types, or to model probabilities of transitions between the six different feather types.

The discovery of multiple types of feather-like structures in pterosaurs has broad implications for our understanding of pterosaur biology and the functional origin of feather-like structures in Avemetatarsalia^{31,32}. Potential functions of these structures include insulation, tactile sensing, streamlining and coloration (primarily for camouflage and signalling), as for bristles, down feathers and mammalian hairs³¹⁻³⁴. Type 1, 2 and 4 filaments could shape a filamentous covering around the body and wings (Fig. 4) that might have functioned in streamlining the body surface in order to

200 reduce drag during flight, as for modern bat fur or avian covert feathers^{33,35}. Type 1
 201 and 2 filaments occur in considerably high densities, particularly around the neck,
 202 shoulder, hindlimb and tail regions where the high degree of superposition prevents
 203 easy discrimination of adjacent fibres. This, along with the wide distribution and
 204 frayed appearance, resembles mammalian underfur adapted for thermal insulation^{36,35}.
 205 Despite the less dense packing of Type 4 filaments on the wings, the morphology of
 206 the structures is consistent with a thermoregulatory function: down feathers can
 207 achieve similar insulation as mammalian hair with only about half the mass, due to
 208 their air-trapping properties and high mechanical resilience, effective in retaining an
 209 insulating layer of still air³⁸. This may optimize the encumbrance of the large wing
 210 area to wing locomotion¹⁸. Type 3 filaments around the jaw (Fig. 4) may have had
 211 tactile functions in e.g. prey handling, information gathering during flight, navigating
 212 in nest cavities and on the ground at night, similar to bristles in birds³⁹.

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Supplementary Information is available in the online version of the paper.

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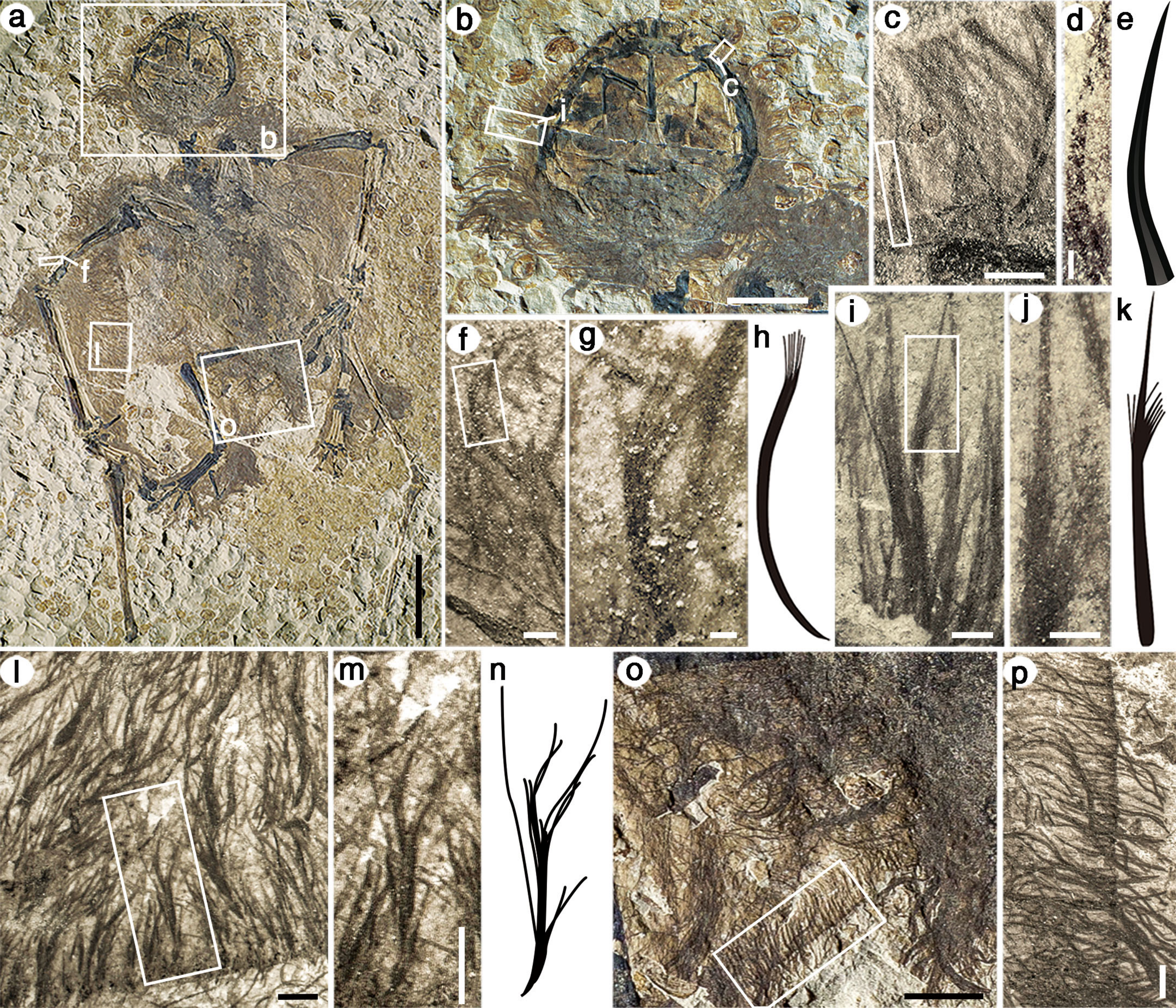
333 **Figure 1 | Integumentary filamentous structures in CAGS–Z070. a**, Overview
 334 shows extensive preservation of soft tissues. **b–p**, Details of the integumentary
 335 filaments in the regions indicated in **a** on the head and neck (**b–d**, **i–j**), forelimb (**f–g**),
 336 wing (**l–m**) and tail (**o–p**), and illustrated reconstructions of the filaments (**e**: Type 1
 337 filament; **h**: Type 2 filament; **k**: Type 3 filament; **n**: Type 4 filament). Scale bars: 20
 338 mm in **a**; 10 mm in **b**; 500 μm in **c** and **i**; 100 μm in **d**; 1 mm in **f**, **l**, **m** and **p**; 200 μm
 339 in **g** and **j**; 5 mm in **o**.

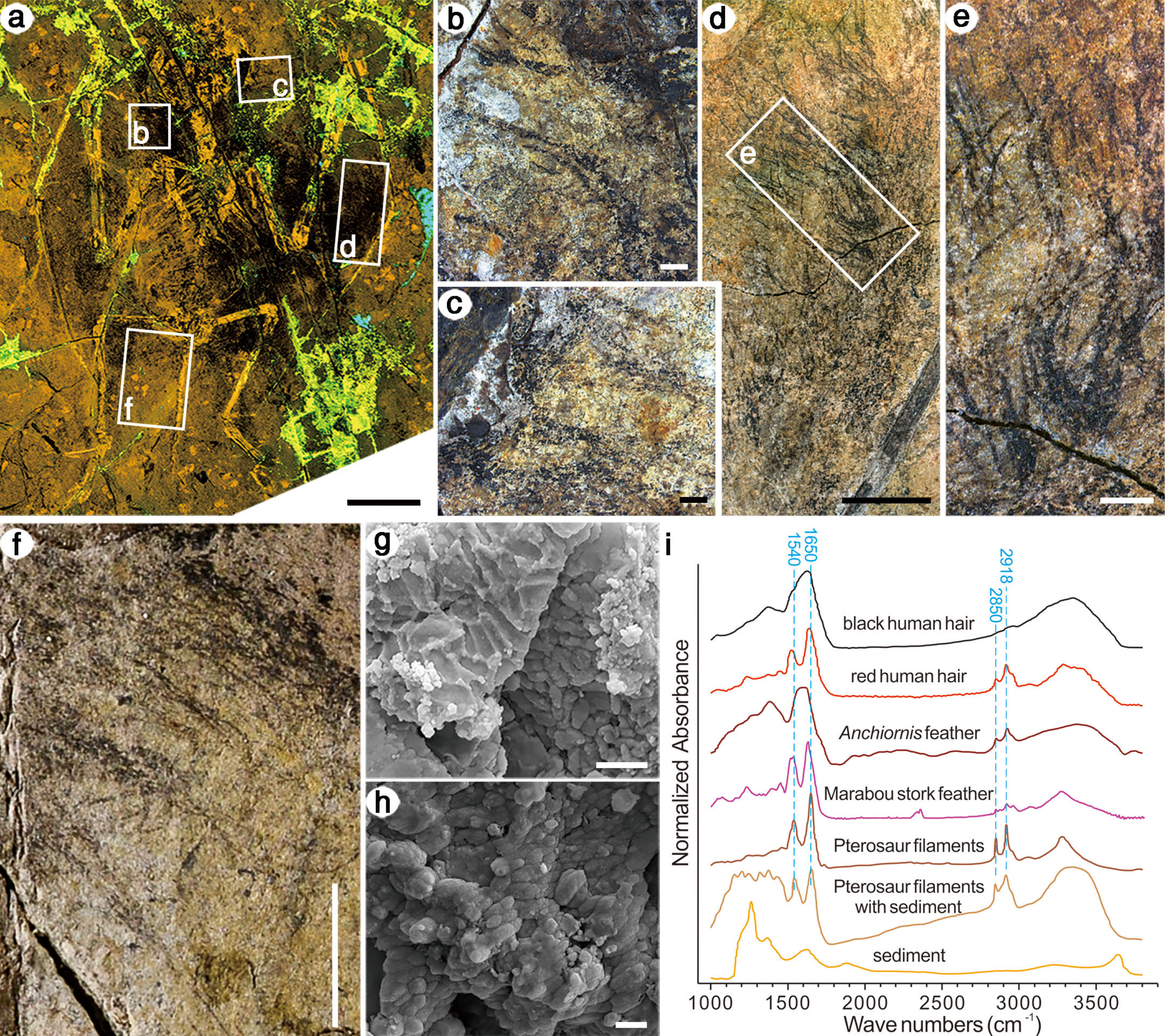
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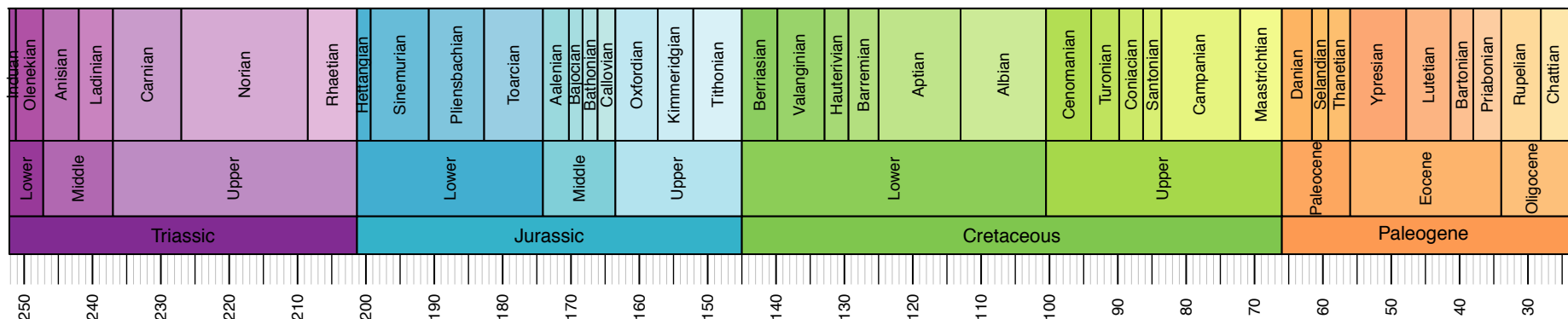
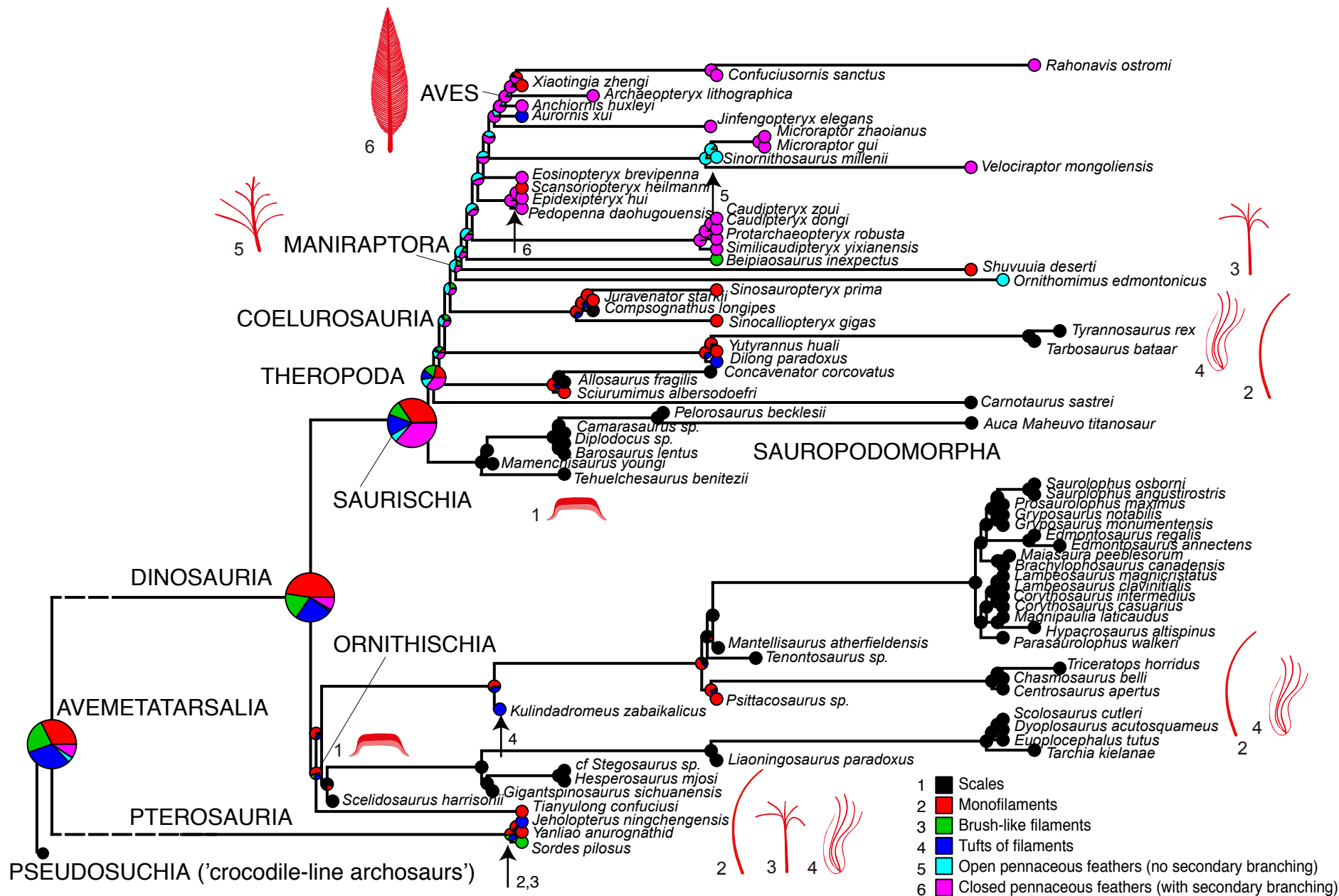
341 **Figure 2 | Preservation, microstructure and chemistry of the integumentary**
 342 **filamentous structures in NJU–57003. a**, Laser-stimulated fluorescence^{6,15,16} image
 343 highlights extensive preservation of soft tissues (black areas). **b–f**, Details of the
 344 integumentary filaments in the regions indicated in **A** on the head and neck (**b–c**),
 345 wing (**d–e**) and tail (**f**). **g–h**, Scanning electron micrographs of the monofilaments on
 346 the neck and hindlimb of NJU–57003 (samples 10 and 39, respectively,
 347 Supplementary Fig. 1a) show densely packed, elongate and oblate melanosomes. **i**,
 348 FTIR absorbance spectra of the monofilaments, monofilaments with sediment matrix,
 349 and sediment matrix in NJU–57003 (Sample 15, Supplementary Fig. 1a) compared
 350 with spectra from a feather of *Anchiornis* (from ref. ²⁰), extant Marabou stork feather
 351 (from ref. ²¹) and black and red human hair melanosomes (from ref. ²⁴). Scale bars: 20
 352 mm in **a**; 1 mm in **b**, **c** and **e**; 5 mm in **d** and **f**; 1 μm in **g** and **h**.

Figure 3 | Phylogenetic comparative analysis of integumentary filament and feather evolution in pterosaurs and archosaurs. The phylogeny is scaled to geological time, with recorded terminal character states for each species, and estimated ancestral character states at the lower nodes. The model is the most likely of the maximum likelihood models, based on minimum-branch lengths (mb1) and transitions occurring as all-rates-different (ARD), but other results with lower likelihoods show scales as ancestral. The ancestral state reconstruction shows a combination of monofilaments, tuft-like filaments, and brush-type filaments as the ancestral state for Avemetatarsalia and for Dinosauria. The estimated ancestral state for Theropoda comprises all five feather states. Numbered small vertical arrows indicate earliest occurrences of feather types 2–6. Two hypotheses for timing of avian feather origins are indicated: A, early origin, at the base of Avemetatarsalia in the Early Triassic, or B, late origin, at the base of Maniraptora in the Early–Middle Jurassic.

Figure 4 | Reconstruction of one of the studied anurognathid pterosaurs, exhibiting diverse types of pycnofibres distributed in different body parts.







A early origin of feathers

B late origin of feathers

